

# Supply Chain Performance Analysis Report

## Executive Summary

This report presents an end-to-end analysis of supply chain delivery performance. A total of **172,765 orders** were analyzed. The findings reveal that **54.71% of orders are delayed**, while only **45.29% are delivered on time**. These delays result in a significant financial impact, with an estimated **\$2.1M loss due to delays**, despite generating a total profit of **\$7.5M**. The analysis identifies operational inefficiencies and key delay drivers across regions, shipping modes, and time periods.

## Dataset Overview

The dataset contains transactional order-level data including shipping timelines, customer segments, product details, and profitability metrics. Data cleaning was performed to remove redundant, irrelevant, and low-variance columns, ensuring high-quality analysis.

## Key Performance Indicators (KPIs)

- Total Orders: 172,765
- Late Deliveries: 94,523
- On-Time Delivery Rate: 45.29%
- Late Delivery Rate: 54.71%
- Total Profit: \$7.5M
- Loss due to Delays: \$2.1M

## Bottleneck Analysis

The analysis highlights that specific shipping modes and regions contribute disproportionately to delivery delays. Certain categories consistently show higher delay percentages, indicating operational bottlenecks in logistics and fulfillment.

## Root Cause Analysis

Key factors contributing to delays include inefficient shipping modes, longer order processing times, and regional logistics challenges. Orders with higher processing times show a strong correlation with delayed deliveries.

## Time-Based Analysis

Delay patterns vary significantly across months, days, and hours. Certain time periods experience higher delays, indicating workload imbalance and operational inefficiencies during peak times.

## Business Recommendations

- Optimize high-delay shipping modes to improve delivery speed
- Reduce order processing time through automation
- Focus on high-delay regions for logistics improvement

- Implement monitoring during peak delay hours
- Use predictive analytics to identify at-risk orders

## **Conclusion**

The supply chain currently faces significant delay challenges affecting both profitability and customer experience. By addressing identified bottlenecks and optimizing operations, the business can improve delivery performance and reduce losses.